MG TECH

PRODUCT CATALOGUE

Liquid PCB Plastic Coating

Electronic Circuit Waterproofing

A heat-free, plastic-based industrial protective coating — a real 20–200 µm layer that makes PCBs, sensors and electronic circuits waterproof and mechanically strong.

mgcoat.com

What it is

Liquid PCB Plastic Coating (MGCoat) is an industrial protective coating that makes printed circuit boards, sensors and electronic circuits waterproof. It is applied at room temperature with no oven or heat and forms a strong, non-transparent, plastic-based layer 20–200 µm thick that protects against water, humidity, washing, condensation and corrosion while adding mechanical strength. It can be removed with a dedicated solvent for repair.

Key advantages

Real waterproofing

Engineered grades from rain-resistant to full immersion, tested powered at 24 V DC and 220 V AC.

Heat-free, room temperature

Applied at 15–45 °C, initial dry in 8–10 min — safe for heat-sensitive boards.

Mechanical reinforcement

A real 20–200 µm plastic layer (not a 1 µm film) that resists handling, vibration and scratches.

Non-transparent coverage

White, black or grey finish hides the sensitive board design while protecting it.

Multi-layer, made to grade

2–4 passes to reach the protection grade and thickness your application needs.

Reworkable

Removable with a dedicated solvent for repair — your board is not locked forever.

Technical specifications

Initial dry time	8–10 minutes
Full cure (max. mechanical strength)	12 hours
Application temperature	15–45 °C (optimal 25–35 °C)
Operating temperature (cured)	−55 °C to +90…110 °C
Thickness per pass	20–200 µm (normal grade 50–100 µm)
Recommended passes	2–4, repeatable to target thickness
Dry time between passes	10 minutes
Coverage	1 L ≈ 1 m^2 at 300–500 µm (more area when thinner)
Colours	White, black or grey (light-transmitting)
Base / finish	Plastic-based, non-transparent
Reworkable	Removable with a dedicated solvent
Adhesion	Film / membrane adhesion
Shelf life	5 years sealed

Protection grades

Environment	Voltage	Passes
Full water immersion	AC 220 V	**4 passes**
Full water immersion	DC up to 24 V	**3 passes**
Rain / splash resistant	AC 220 V	**2 passes**
Rain / splash resistant	DC up to 24 V	**2 passes**

10 minutes initial dry are required between passes.

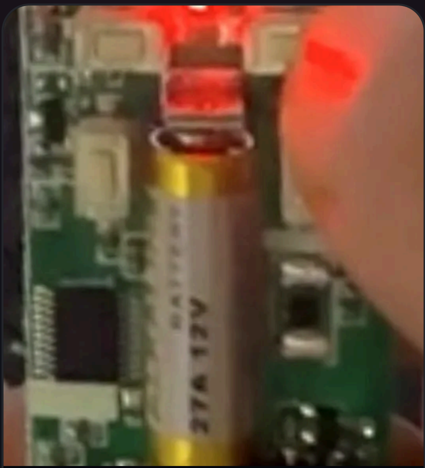
How it is applied

1. Clean the board and let it dry fully.
2. Mask connectors and any areas that must stay exposed.
3. Apply by full dip (or spray / brush). Make sure every part is covered with no trapped air.
4. Wait 10 minutes initial dry, then repeat — 2 to 4 passes per the required protection grade.
5. Visually inspect: the whole board must be covered, with no air pockets.
6. Full cure for 12 hours for maximum mechanical strength, then test.

Verified tests

Powered immersion test	AC & DC, 27 V and 220 V — continuous operation
Humidity test	+95 % RH — continuous operation
Salt-spray test	Continuous operation
Operating range	−55 °C to +110 °C
Durability	The cured plastic film keeps protecting until it meets heat above 120 °C or a hard scratch (drill / cutting blade).

Real test


1. PCB board


2. Submerged in water after coating


3. Powered and working under water

Where to use it

- Automotive ECUs & modules
- Motorcycle electronics
- Drones & FPV
- CCTV & outdoor cameras
- Marine sensors & boat lights
- EV & battery management (BMS)
- Solar inverters
- LED boards & signage
- Repair workshops

Why MGCoat

Many market coatings are ultra-thin transparent films that add no mechanical protection. MGCoat builds a real plastic layer — non-transparent, reinforced, immersion-capable and reworkable — tested powered under water at 24 V DC and 220 V AC. It is a stronger, more durable choice for harsh, outdoor and high-reliability environments.

Packaging & pricing

1-5 L	Standard packaging
50 L	+ 10 L free
100 L	+ 24 L free

Up to 5 L	**$199 — Turkey only**
50 L and above	**$124 / unit — export packaging**

Pricing and specifications are confirmed at the time of order.

Safety & handling

- Plastic-based: like most plastics it is somewhat flammable in liquid form — never expose the liquid to open flame.
- Once cured it is a stable plastic film and keeps working up to 120 °C.
- Reworkable: removable with a dedicated solvent for repair.
- After coating, always visually confirm full, air-free coverage of the board.

Order & contact

Get a quote or place an order — message us directly on WhatsApp. Available in English, Russian, Turkish, Arabic and Persian. We also welcome resellers and regional partners.

Order on WhatsApp

WhatsApp +90 552 876 7973 · Instagram @mgcoatcom · mgcoat.com